Experimental evaluation of quantum physically unclonable functions on cloud-accessible quantum processors

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Quantum physically unclonable functions (QPUFs) promise device authentication whose security rests on quantum-mechanical rather than computational assumptions. Their viability on noisy intermediate-scale quantum (NISQ) hardware has not been tested systematically. We implement a phase-estimation-based QPUF (PE-QPUF) as a two-stage quantum phase estimation (QPE) protocol with mid-circuit measurement and run it on trapped-ion and superconducting processors accessed through Amazon Braket. Haar-random single-qubit unitaries serve as the device secret and are compiled to native gates by Euler decomposition. For multi-qubit targets, powers of the unitary are computed classically and re-projected onto the unitary group by singular value decomposition, so that the number of controlled gates grows only linearly with the number of precision qubits. GHZ-state, quantum Fourier transform (QFT) and QPE benchmarks relate the observed failures to connectivity, SWAP overhead and per-task gate limits. On IonQ Forte-1 with one target qubit and up to ten precision qubits, the coincidence counts of the two phase-estimation outcomes form two sharp clusters, and the acceptance rate at small tolerance exceeds the uniform-guess probability by a factor of 2.6 to 140. With two and three target qubits and 10 to 15 precision qubits, the outcomes are uncorrelated and the acceptance follows the random-guess line. We discuss error mitigation for this protocol and the hardware features that decide whether a QPUF can be demonstrated on a given processor.

I. INTRODUCTION

Quantum processors based on superconducting circuits, trapped ions, neutral atoms and photons are now offered through cloud services. They differ widely in qubit number, connectivity, gate fidelity and supported operations. Application-level tests complement standard device metrics by showing how a structured task behaves on real hardware [1, 2].

Physically unclonable functions (PUFs) are physical systems whose uncontrollable fabrication variations act as a fingerprint, so that a device can be authenticated by challenge-response measurements [3–5]. Quantum PUFs generalize this idea: a QPUF is an unknown, physically unclonable unitary or channel that is queried with quantum states. Formal definitions, security notions and impossibility results were given by Arapinis *et al.* [6], identification protocols by Doosti *et al.* [7], and efficient constructions from unitary t-designs by Kumar *et al.* [8]. In the PE-QPUF protocol the verifier checks the consistency of two successive phase-estimation outcomes obtained from the same unknown unitary. A tolerance Δ on their cyclic distance trades robustness against forgery security.

The protocol is simple on paper but demanding in practice. It requires Haar-random unitaries, controlled powers U^{2^k} on many precision qubits, mid-circuit measurement, and enough coherence to carry a token state between two QPE rounds. We therefore pursue two goals: to test whether a QPUF can be authenticated on real devices, and to use it as a structured benchmark that exposes the practical constraints of cloud-accessible processors. Our main results are as follows. (i) We give a hardware-oriented implementation of the two-stage PE-QPUF, including compilation of Haar-random unitaries and their controlled powers (Sec. III). (ii) We benchmark IonQ, IQM and Rigetti devices with GHZ, QFT and QPE circuits (Sec. IV). (iii) We report authentication experiments on IonQ Forte devices with one to three target qubits and up to 15 precision qubits (Sec. V). (iv) We assess error mitigation and a first superconducting run (Secs. VI and VII).

II. BACKGROUND

A. Quantum phase estimation

For a unitary U with eigenstate $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, QPE uses d precision qubits in $|+\rangle^{\otimes d}$, controlled-U^{2^k} operations for $k = 0, \ldots, d-1$, and an inverse QFT, and outputs an integer $m \approx \varphi 2^d$ [9–11]. If the target is not an eigenstate, the outcome m projects it approximately onto an eigenspace of U. QPE therefore acts as an approximate von Neumann measurement of U, which is the property that the QPUF uses.

B. PE-QPUF authentication

The *authenticator* holds the secret unitary U and the *verifier* checks its identity.

1. *Token generation.* QPE with precision register P_1 is run on a fixed target state, and P_1 is measured, giving m_1. The target collapses to an approximate eigenstate of U with phase $m_1/2^d$ and serves as the token.

2. *Verification.* A second QPE with register P_2 acts on the token and gives m_2. Ideally $m_2 = m_1$.

3. *Decision.* The device is accepted if the cyclic distance $d_c(m_1, m_2) = \min(|m_1 - m_2|, 2^d - |m_1 - m_2|) \leq \Delta$.

A larger Δ raises the acceptance probability of the legitimate device (completeness) but also the success probability of an adversary who does not hold U (soundness). A uniformly random guess is accepted with probability $(2\Delta+1)/2^d$, which is the baseline for all hardware results below. The tolerance is bounded by 2^{d-1} [6].

C. Haar-random secret

The unknownness of an ideal QPUF requires U to be Haar random [6, 8]. Exact sampling on n qubits needs exponentially large circuits, whereas unitary t-designs give efficient approximations [12]. For $n = 1$ we sample exactly. Up to a global phase,

$$U = R_z(\beta)R_y(\gamma)R_z(\delta), \tag{1}$$

with $\beta, \delta \sim \text{Uniform}[0, 2\pi)$ and $\cos\gamma \sim \text{Uniform}[-1, 1]$. For $n \geq 2$ we use the QR decomposition of a complex Ginibre matrix with the phase correction of Mezzadri [13].

III. IMPLEMENTATION

A. Circuit

The two-stage protocol is a single Qiskit circuit [14]. Register P_1 is measured mid-circuit and the target is left in its post-measurement state, and a second QPE with P_2 follows. Each shot collapses independently, so the sequence is a genuine sequential protocol. A variant without mid-circuit measurement would run the two QPE circuits coherently and would not test the collapse. The seed of every unitary is logged, which allows noiseless local simulation of each hardware run with the same unitary. We do not model storage of the token in quantum memory between the stages, so the experiment tests the sequential protocol and circuit noise, not storage decoherence.

B. Controlled powers

Applying controlled-U a total of 2^k times is infeasible for $d \gtrsim 10$. We compute U^{2^k} classically by repeated squaring and compile it directly. For one target qubit, U^{2^k} is a 2×2 unitary whose Euler angles give a controlled gate made of three controlled rotations, independent of k, so the gate count is linear in d. For several target qubits the matrix is passed to Qiskit's `UnitaryGate` and decomposed with a quantum Shannon or KAK-type decomposition [15, 16]; this lowers the gate count but the compilation time grows rapidly with the number of target qubits.

For $d \approx 30$ repeated squaring accumulates errors of order $2^k\epsilon_{\text{machine}} \approx 10^{-6}$ (with $\epsilon_{\text{machine}} \approx 10^{-16}$), which exceeds Qiskit's unitarity tolerance of 10^{-8}. We re-project each power onto the unitary group through the singular value decomposition $U = W\Sigma V^\dagger \mapsto WV^\dagger$. All singular values of a unitary equal one, so the projection is exact in infinite precision and removes the drift.

C. Native gates and workflow

Circuits are transpiled to $\{R_z, R_x, R_{xx}, \text{measure}\}$ before submission to IonQ, which compiles to GPI, GPI2 and Mølmer–Sørensen gates [17] with no significant change of the gate count. On IQM and Rigetti devices the compiler inserts SWAP gates for non-adjacent qubits. A submission script returns a task identifier, a second script retrieves completed results and logs all metadata, and analysis notebooks compare hardware outcomes with the local simulation.

IV. HARDWARE BENCHMARKING

A. Devices

Table I lists the devices considered. Prices and availability are those at the time of the experiments (April to November 2026) and change frequently [18]. IonQ Forte has a median two-qubit fidelity of about 99.6% and a state-preparation and measurement error of about 0.5%, with built-in debiasing and sharpening (minimum 2500 shots per mitigated task). IQM Garnet has median fidelities of 99.92% (one qubit) and 99.51% (two qubits). Rigetti Cepheus-1-108Q is a modular array of twelve 9-qubit chiplets with a median two-qubit fidelity of 99.1%. IonQ Aria-1/2 and Rigetti Ankaa-3 were retired during the study and are excluded.

B. Benchmark circuits

GHZ state. An n-qubit GHZ state [19] needs one Hadamard and $n-1$ CNOT gates. The metric is the ideal fraction

$$F_{\text{GHZ}} = P(0\cdots0) + P(1\cdots1), \tag{2}$$

which is 1 for a noiseless device and approaches $2/2^n$ for a fully depolarized one.

QFT. The QFT applied to $|0\rangle^{\otimes n}$ should give a uniform distribution. We quantify the deviation by the total variation distance $\text{TVD} = \frac{1}{2}\sum_k |P_{\text{obs}}(k) - 2^{-n}|$. For N shots its noiseless value is not zero, and for $N \ll 2^n$

TABLE I. Devices on Amazon Braket considered in this study. Every task carries a fixed fee of 0.30 USD. MCM denotes mid-circuit measurement. QuEra Aquila supports only analog Hamiltonian simulation and cannot run the circuits used here.

Provider	Device	Technology	Qubits	Connectivity	MCM	Shot price (USD)
IonQ	Forte-1, Forte-Enterprise-1	Trapped $^{171}Yb^{+}$	36	All-to-all	Yes (no reset)	0.08
AQT	IBEX-Q1	Trapped $^{40}Ca^{+}$	12	All-to-all	Not documented	0.0235
IQM	Garnet	Superconducting	20	Square lattice	Experimental	0.00145
IQM	Emerald	Superconducting	54	Square lattice	Experimental	0.0016
Rigetti	Cepheus-1-108Q	Superconducting	108	Square lattice	Not documented	Not listed
QuEra	Aquila	Neutral atoms (analog)	≤ 256	Configurable	n/a	0.01

TABLE II. GHZ ideal fraction F_{GHZ}. A value of 0.000 means that no shot returned either ideal bitstring.

Device	n	F_{GHZ}
IonQ Forte-Enterprise-1	18	0.711
IonQ Forte-Enterprise-1	36	0.465
IQM Garnet	10	0.531
IQM Garnet	20	0.074
Rigetti Ankaa-3	41, 82	0.000
Rigetti Cepheus-1-108Q	54	0.000

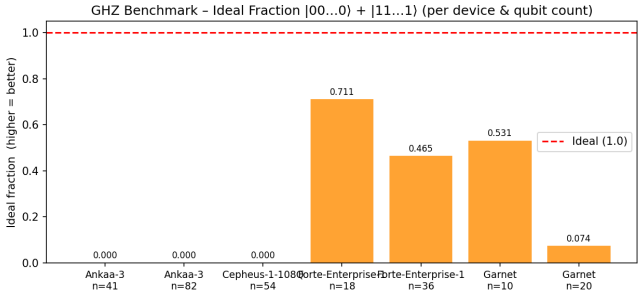


FIG. 1. GHZ ideal fraction F_{GHZ} for each device and qubit count. The dashed line marks the ideal value 1.

it approaches $1 - N/2^n$, so large n with few shots gives TVD ≈ 1 regardless of device quality. The simulator with 1024 shots gives TVD ≈ 0.13 at $n = 6$ as a finite-sampling reference.

QPE of the T gate. For $T = \mathrm{diag}(1, e^{i\pi/4})$ acting on $|1\rangle$, $\varphi = 1/8$, so the ideal outcome is $m^* = 2^n/8$ and the metric is $A_{\mathrm{peak}} = P(m^*)$.

C. Results

Table II and Fig. 1 show the GHZ results. The simulator gives $F_{\mathrm{GHZ}} \approx 1$ at $n = 6$. IonQ Forte-Enterprise-1 keeps substantial GHZ fidelity up to its full 36 qubits, consistent with all-to-all connectivity and low two-qubit error. IQM Garnet drops from 0.531 at $n = 10$ to 0.074 at $n = 20$, which we attribute to SWAP routing on the square lattice. For both Rigetti devices the output is flat over all bitstrings.

On IQM Garnet the QFT gives TVD = 0.787 at $n = 10$ and 1.000 at $n = 20$. At $n = 20$ the histogram is flat at about 1/256 because only 256 shots are spread over 2^{20} outcomes, so TVD = 1 reflects undersampling and shows that the output is noise dominated; it does not by itself mean that all shots fell on one bitstring. For QPE, Garnet gives $A_{\mathrm{peak}} = 0.012$ at $n = 10$ and 0.000 at $n = 20$, and the most probable outcome at $n = 10$ is not the ideal bin. [Verify: re-run the noiseless simulator for the T-gate benchmark with an explicit eigenstate input; the ideal A_{peak} must be 1 for an exact eigenphase, whereas the earlier simulation gave about 0.15 at $n = 5$. State the ideal bin for each n.]

Role of the two-qubit gate count. Trapped-ion devices keep QPE performance to larger registers than superconducting devices. A crude estimate shows why. A transpiled circuit with N two-qubit gates of fidelity f has fidelity of order f^N. Routing a gate between non-adjacent qubits costs SWAP gates of three CNOTs each, and the overhead is quoted as a factor 3 to 10 for arbitrary-unitary circuits on square lattices. With $f = 0.99$ and $N \approx 500$, $f^N \approx 0.0066$. For IonQ Forte with a two-qubit infidelity of 0.75%, $f^N \approx 1/e$ after about 133 gates, so a few hundred two-qubit gates are the effective limit. This agrees with the guidance from the Braket technical staff that a 20 000-gate circuit is unlikely to give meaningful results.

D. Platform constraints

Several service limits restrict which circuits can run, independently of hardware quality.

- *Gate and shot limits.* On-demand IonQ tasks are limited to 10^6 as the product of gate number and shot number. This permitted 500 shots up to six precision qubits and 400 shots at seven; above this the shot number had to be lowered by roughly 100 per additional qubit (100 shots at $d = 10$). Braket Direct lifts this limit but enforces a maximum of 5000 shots per task on IonQ devices.

- *Circuit size.* Task creation fails above 20 000 gates.

A 54-qubit QPE on Cepheus-1-108Q has 43 622 gates and the 54-qubit QFT has 44 688, so neither could run. Outside IQM Garnet we could not run the QFT on even half of a device's qubits.

- *Qubit count.* A request for all 108 qubits of Cepheus-1-108Q was rejected because 107 were reported available, and a 108-qubit GHZ task failed without an error message. Device properties should be queried before submission.

- *Connectivity.* On Garnet with one target qubit, the nominal limit of nine precision qubits could not be reached because buffer qubits are needed for SWAP routing.

- *Mid-circuit measurement* is required by the QPUF. It is available on IonQ Forte devices, but reset is not supported on Forte-Enterprise-1, which prevents qubit reuse.

V. QPUF EXPERIMENTS

A. One target qubit on IonQ Forte-1

With one target qubit the secret is compiled with three controlled rotations per precision qubit. We ran $d = 4, 5, 6, 7, 9$ and 10 with 500 shots for $d \leq 6$, 400 at $d = 7$, 200 at $d = 9$ and 100 at $d = 10$, as the gate-shot limit required. Each hardware run is paired with a noiseless simulation of the same unitary regenerated from the logged seed.

The raw output is a list of pairs (m_1, m_2), where m_1 is read from the first precision register and m_2 from the second. Figures 2 and 3 show the number of shots at each pair as the area of a circle. For a noiseless device and an exact eigenstate token, all shots lie on the diagonal $m_1 = m_2$. A count on the diagonal at an eigenphase position is a *coincidence*: the second QPE reproduced the integer of the first. A count on a horizontal or vertical line through a cluster means that one readout moved while the other stayed.

Two coincidence clusters at $\pm\varphi$. Every map has two clusters, at (m_+, m_+) and (m_-, m_-), which sharpen with increasing d. Their positions, read from Figs. 2 and 3, scale with 2^d and satisfy $m_+ + m_- = 2^d$ (Table III). This is expected, since the Euler form of Eq. (1) omits the global phase, so $U \in \mathrm{SU}(2)$ has eigenvalues $e^{\pm 2\pi i\varphi}$. In each shot the first QPE collapses the target onto one of the two eigenstates, and the second QPE returns the same integer. Each cluster collects about half of the counts.

Acceptance rate. The acceptance rate at tolerance Δ is the fraction of shots with $d_c(m_1, m_2) \leq \Delta$, i.e. the cumulative coincidence count along the diagonal. Table IV gives the values at $\Delta = 0$ and 2 and Fig. 4 the full curves.

Three observations follow from the coincidence counts. (i) The simulator is below 1 at small Δ. At $\Delta = 0$ the noiseless value is only 0.51 to 0.59, because QPE with finite d is an approximate measurement: the eigenphase is not an integer multiple of 2^{-d} and sidelobes spread the counts. This is the intrinsic completeness error of the protocol, so the hardware noise is the gap to the simulator, not the distance to 1. (ii) The gap grows with d. At $\Delta = 2$ the deficit is 0.06 at $d = 4$, 0.21 at $d = 7$ and 0.25 at $d = 10$, with one-standard-deviation statistical errors of about 0.02, 0.02 and 0.05. The growth follows the gate count, which rises with d through the controlled rotations and the $O(d^2)$ inverse QFT. (iii) The authenticator stays far above guessing. At $\Delta = 2$ the acceptance exceeds the uniform-guess probability by a factor of about 2.6 at $d = 4$, 17 at $d = 7$ and 140 at $d = 10$. A uniform guess is a weak adversary, since an attacker who knew the two cluster positions would do better, so a security statement must use the forgery bound of Ref. [6].

TABLE III. Cluster positions read from the hardware coincidence maps. They agree with the simulator in every case. [Confirm that one seed was used for all d.]

d	(m_+, m_-)	$m_+ + m_-$	$m/2^d$	
4	4, 12	16	0.25	0.75
5	9, 23	32	0.28	0.72
6	17, 47	64	0.27	0.73
7	34, 94	128	0.27	0.73
10	≈275, ≈750	≈1025	≈0.27	≈0.73

At large Δ the $d = 10$ hardware curve climbs in steps to 0.91 at $\Delta \approx 250$ and reaches 1 only at $\Delta = 2^{d-1} = 512$. These tail counts come from shots in which one readout fell on the wrong cluster or on a random integer, and no practical Δ should accept them.

B. Several target qubits under Braket Direct

Under a Braket Direct reservation on IonQ Forte-Enterprise-1 the gate limit does not apply, and we ran two and three target qubits with 10 to 15 precision qubits (1000 shots, and 800 shots for three targets), compiling the matrix powers as in Sec. III.

Here the pairs (m_1, m_2) no longer form clusters. Figure 5 shows each run as a histogram over m_1 and $m_1 - m_2$ and as the acceptance against Δ. A working QPUF gives a ridge at $m_1 - m_2 = 0$ that is highest at the eigenphases of the secret unitary. For $(d, n_t) = (10, 2)$, $(15, 2)$ and $(10, 3)$ the histogram is a flat field. The acceptance curves are straight lines reaching 1 at $\Delta \approx 2^{d-1}$ (about 5×10^2 for $d = 10$ and 1.6×10^4 for $d = 15$), which is the curve $(2\Delta + 1)/2^d$ for two independent uniform integers, and the acceptance at $\Delta = 2$ is close to zero. These runs therefore carry no authentication signal. This agrees with the gate budget: each controlled multi-qubit unitary needs many two-qubit gates, and for an infidelity of 0.75% the output becomes indistinguishable from noise

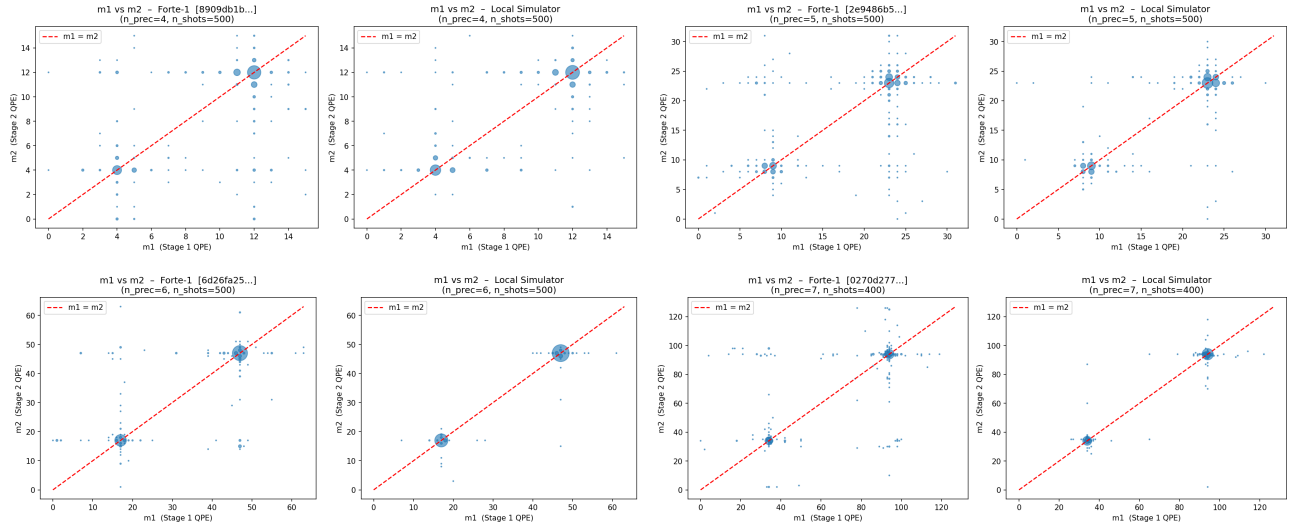


FIG. 2. Coincidence maps (m_1 against m_2) of the two-stage QPUF on IonQ Forte-1 (left panel of each pair) and the noiseless simulator (right panel) for $d = 4, 5, 6$ and 7 precision qubits (500, 500, 500 and 400 shots). Circle area is proportional to the number of shots and the dashed line is $m_1 = m_2$.

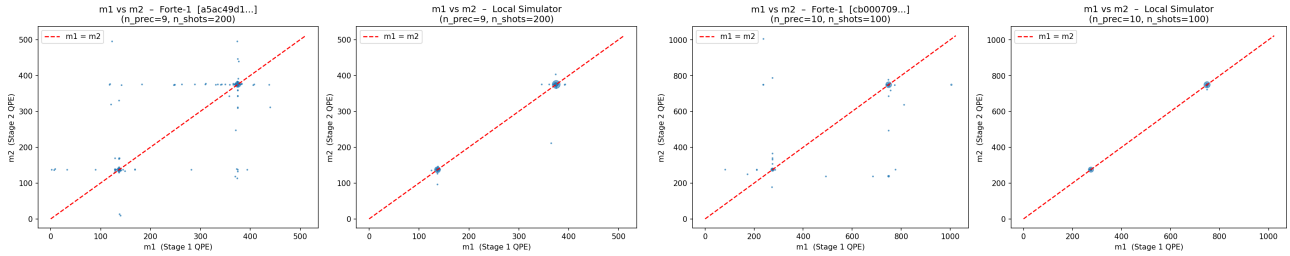


FIG. 3. As Fig. 2, for $d = 9$ (200 shots) and $d = 10$ (100 shots).

after roughly 10^2 gates. [Insert transpiled two-qubit gate counts, estimated circuit fidelity f^N and wall-clock time per task for each configuration.]

VI. ERROR MITIGATION

Error correction would suppress noise but needs many extra qubits and gates, and each extra gate is itself a noise source. Error mitigation instead post-processes results from noisy circuits, at the cost of additional executions [20–23]. We consider (i) zero-noise extrapolation (ZNE) by digital gate folding [20, 24], (ii) readout-error mitigation [25], and (iii) IonQ's debiasing and sharpening.

The QPUF poses two specific issues. First, the relevant quantity is the joint distribution of discrete outcomes (m_1, m_2) per shot, not an expectation value, so ZNE must be applied to the derived acceptance probability $P_{\mathrm{acc}}(\Delta)$ and not to single shots. Second, the mid-circuit collapse prevents any technique that re-runs the same quantum state. We therefore estimate $P_{\mathrm{acc}}(\Delta)$ at noise scales $\lambda \in \{1, 3, 5\}$, extrapolate to $\lambda = 0$, and compare with the noiseless simulation and the random-guess baseline.

[Insert error-mitigation results: (i) a table of device, d, n_{t}, shots, raw, mitigated, noiseless and baseline $P_{\mathrm{acc}}(\Delta = 2)$; (ii) extrapolated acceptance curves with confidence bands; (iii) the number of additional executions and cost; (iv) the effect on the forgery probability of a random-guessing adversary, which must stay at the baseline.]

VII. SUPERCONDUCTING HARDWARE

To study a superconducting architecture we ran the QPUF on Rigetti Cepheus-1-108Q within a one-hour Braket Direct reservation on 10 August 2026, together with the error mitigation of Sec. VI. The square-lattice topology requires extra SWAP gates, so a noisier output is expected than on all-to-all trapped-ion devices. Earlier on-demand tests support this: QPE on 20 qubits ran but gave a poor result, and an acceptable result at five precision qubits degraded at seven and nine. [Insert results of the reservation: transpiled gate count with and without routing, acceptance rate, and mitigated acceptance rate.]

TABLE IV. Acceptance rate on IonQ Forte-1 and in the noiseless simulator. Values are read from the plotted curves (±0.02). [Replace with exact values from the job logs.]

d	Shots	Δ	Hardware	Simulator	Uniform guess $(2\Delta+1)/2^d$
4	500	0	0.48	0.55	0.06
4	500	2	0.81	0.87	0.31
7	400	0	0.35	0.51	0.008
7	400	2	0.66	0.87	0.039
10	100	0	0.41	0.59	0.001
10	100	2	0.68	0.93	0.005

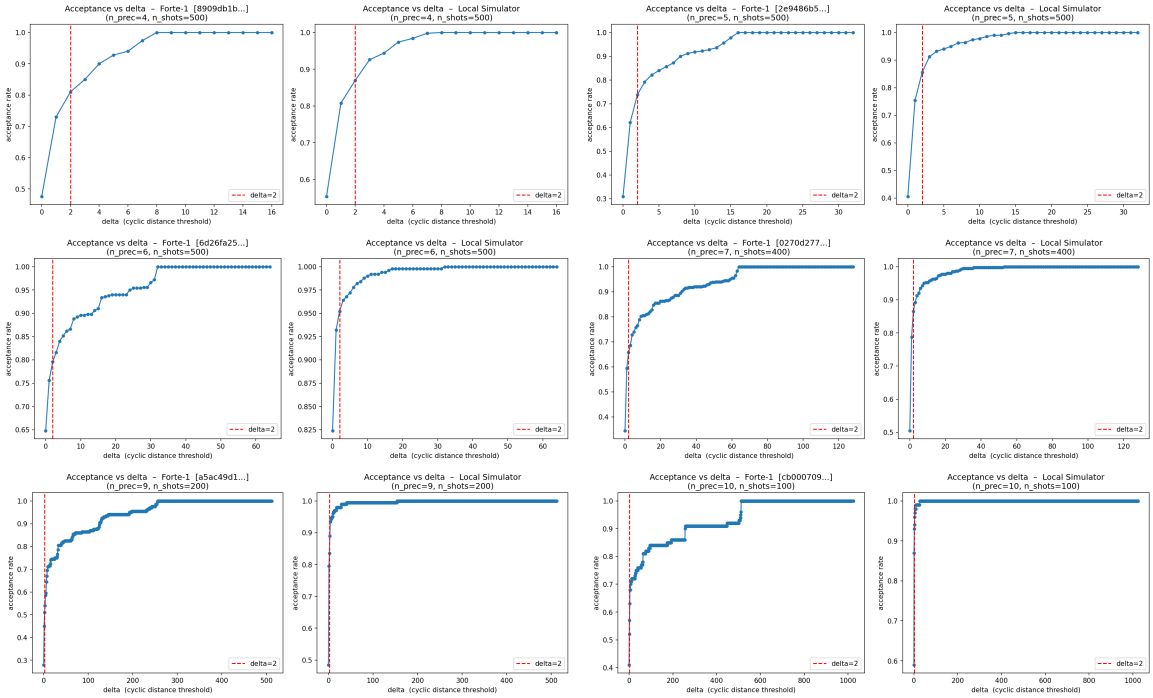


FIG. 4. Acceptance rate against Δ for $d = 4, 5$ (top), $6, 7$ (middle) and $9, 10$ (bottom). Each pair shows the hardware (left) and the noiseless simulator (right). The dashed line marks $\Delta = 2$.

VIII. DISCUSSION

The protocol needs mid-circuit measurement, a deep circuit with controlled powers on d precision qubits, and for $n_t > 1$ controlled n-qubit unitaries of exponential complexity. Mid-circuit measurement is met on trapped-ion devices, whereas the second and third requirements are limited by an effective budget of 10^2 to 10^3 two-qubit gates. Authentication works for one target qubit and up to ten precision qubits. This regime is too small for cryptographic security, since the guessing probability $(2\Delta+1)/2^d$ is only about 5×10^{-3} at $\Delta = 2$ and $d = 10$, and larger d and n_t move the circuit into the noise-dominated regime.

The tolerance Δ must exceed the hardware-induced spread of $m_2 - m_1$ and stay below the value at which guessing succeeds with non-negligible probability; for a noisy device this window may be empty. Error mitigation can widen the window for the legitimate device but cannot reduce the adversary's baseline.

For this task the ranking of devices follows from mid-circuit measurement support, connectivity, two-qubit error and the cap on gates per task. Trapped ions satisfy these criteria best, at a shot price 50 to 90 times above superconducting devices and with longer execution times.

Limitations. The token is not stored in quantum memory between the stages, and no adversarial strategy such as approximate cloning has been tested. The Haar-random secret is exact only for $n_t = 1$. Runs used 100 to 1000 shots, so statistical uncertainty on small acceptance rates is not negligible, and device specifications and prices change frequently.

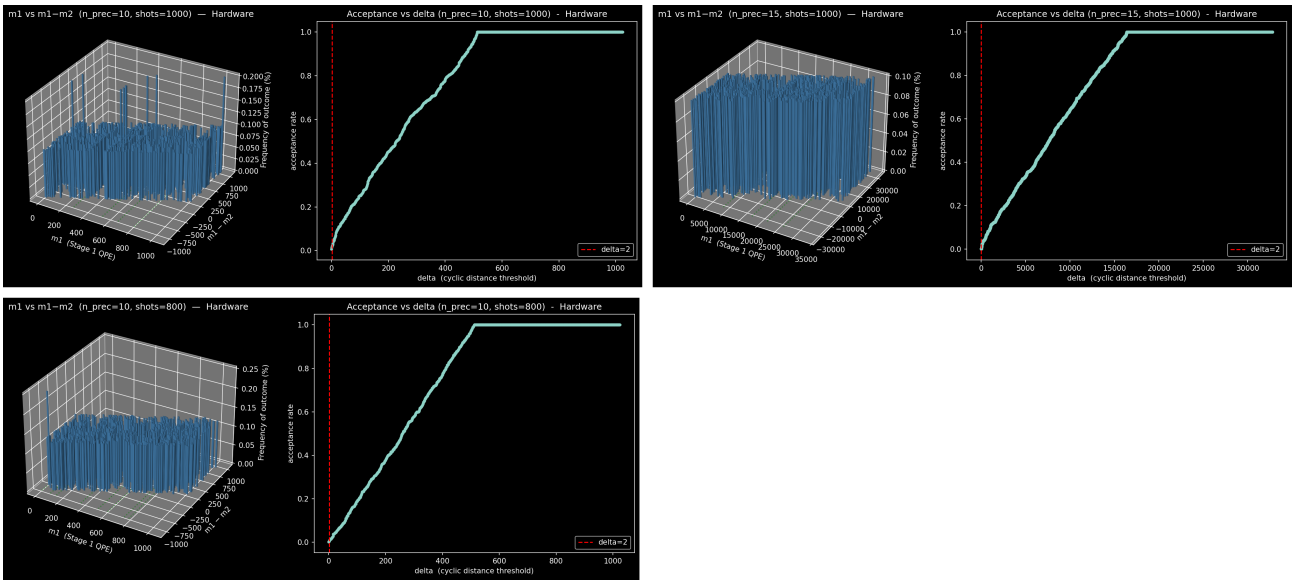


FIG. 5. Multi-target QPUF on IonQ Forte-Enterprise-1. Top left: $d = 10$, $n_t = 2$ (1000 shots). Top right: $d = 15$, $n_t = 2$ (1000 shots). Bottom left: $d = 10$, $n_t = 3$ (800 shots). In each case the left panel is the frequency of outcomes over m_1 and $m_1 - m_2$ and the right panel the acceptance against Δ; the dashed line marks $\Delta = 2$.

IX. CONCLUSION

We implemented a sequential PE-QPUF with mid-circuit measurement and ran it on cloud-accessible processors. For one target qubit and up to ten precision qubits, IonQ Forte-1 produces sharp coincidence clusters at $\pm\varphi$ and accepts the device with high probability at small Δ. For multi-qubit targets with 10 to 15 precision qubits the outcomes are uncorrelated and the acceptance equals the random-guess line. GHZ, QFT and QPE benchmarks show that all-to-all connectivity and mid-circuit measurement are the decisive hardware features for this protocol class, and that platform limits on gates and shots restrict the usable part of the advertised qubits. Future work includes quantum memory for the token, approximate unitary designs for multi-qubit secrets, and adversarial tests of unforgeability.

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